

1. Use Part 1 of the Fundamental Theorem of Calculus to find the derivative of each of the following functions.

$$1). g(x) = \int_1^x \frac{1}{t^4 + 2} dt.$$

$$g'(x) = \frac{1}{x^4 + 2}$$

$$2). g(x) = \int_{x^2}^1 \cos \sqrt{t} dt.$$

$$= \int_1^{x^2} \cos \sqrt{t} dt$$

$$g'(x) = -\cos \sqrt{x^2} \cdot 2x$$

$$= -2x \cos x$$

$$3). g(x) = \int_{-x}^{\cos x} (1+t^2)^5 dt$$

$$= \int_{-x}^0 (1+t^2)^5 dt + \int_0^{\cos x} (1+t^2)^5 dt$$

$$= -\int_0^{-x} (1+t^2)^5 dt + \int_0^{\cos x} (1+t^2)^5 dt$$

$$g'(x) = -(1+(-x)^2)^5 \cdot (-1) + (1+(\cos x)^2)^5 \cdot (-\sin x)$$

$$= (1+x^2)^5 - \sin x \cdot (1+\cos^2 x)^5$$

2. Evaluate the integral. (Note: Use the Fundamental Theorem of Calculus-Part 2)).

$$1). \int_{\pi}^{2\pi} \tan \theta \cos \theta d\theta$$

$$= \int_{\pi}^{2\pi} \frac{\sin \theta}{\cos \theta} \cos \theta d\theta$$

$$= \int_{\pi}^{2\pi} \sin \theta d\theta$$

$$= -\cos \theta \Big|_{\pi}^{2\pi}$$

$$= -\cos(2\pi) - (-\cos \pi)$$

$$= -1 + (-1) = -2$$

$$3). \int_{\pi/6}^{\pi/4} \sec^2 x dx$$

$$= \tan x \Big|_{\pi/6}^{\pi/4}$$

$$= \tan\left(\frac{\pi}{4}\right) - \tan\left(\frac{\pi}{6}\right)$$

$$= 1 - \frac{1}{\sqrt{3}}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}}$$

$$2). \int_1^9 \frac{x-1}{\sqrt{x}} dx$$

$$= \int_1^9 \left(\frac{x}{\sqrt{x}} - \frac{1}{\sqrt{x}} \right) dx = \int_1^9 (x^{1/2} - x^{-1/2}) dx$$

$$= \left(\frac{2}{3} x^{3/2} - \frac{1}{1/2} x^{1/2} \right) \Big|_1^9$$

$$= \left[\frac{2}{3} (\sqrt{x})^3 - 2\sqrt{x} \right] \Big|_1^9 = \left[\frac{2}{3} (3^3) - 2 \cdot 3 \right] - \left(\frac{2}{3} - 2 \right)$$

$$= 18 - 6 - \frac{2}{3} + 2 = \frac{40}{3}$$

$$4). \int_1^{\sqrt{3}} \frac{4}{t^2 + 1} dt$$

$$= 4 \arctan t \Big|_1^{\sqrt{3}}$$

$$= 4 \arctan(\sqrt{3}) - 4 \arctan(1)$$

$$= 4 \cdot \frac{\pi}{3} - 4 \cdot \frac{\pi}{4}$$

$$= \frac{\pi}{3}$$

$$5). \int_{-1}^2 (x^3 - 2x) dx$$

$$= \left(\frac{1}{4} x^4 - x^2 \right) \Big|_{-1}^2$$

$$= \left(\frac{1}{4} \cdot 16 - 4 \right) - \left(\frac{1}{4} \cdot (-1)^4 - (-1)^2 \right)$$

$$= (4 - 4) - \left(\frac{1}{4} - 1 \right)$$

$$= \frac{3}{4}$$

$$6). \int_{-1}^1 e^{u+1} du$$

$$= \int_{-1}^1 e \cdot e^u du$$

$$= e \cdot e^u \Big|_{-1}^1$$

$$= e \cdot (e - e^{-1})$$

$$= e^2 - 1$$

$$7). \int_0^2 (2 + x^5)^2 dx$$

$$= \int_0^2 (4 + 4x^5 + x^{10}) dx$$

$$= \left(4x + \frac{4}{6} x^6 + \frac{1}{11} x^{11} \right) \Big|_0^2$$

$$= 8 + \frac{2}{3} \cdot 2^6 + \frac{1}{11} \cdot 2^{11}$$

$$= 8 + \frac{128}{3} + \frac{2048}{11}$$

$$= \frac{7816}{33}$$

$$8). \int_0^2 2^x dx$$

$$= \frac{1}{\ln 2} \cdot 2^x \Big|_0^2$$

$$= \frac{1}{\ln 2} (2^2 - 2^0)$$

$$= \frac{3}{\ln 2}$$

3. Evaluate $\int_{-2}^2 f(x) dx$, where $f(x) = \begin{cases} 3, & x \geq -1 \\ \frac{1}{x}, & x < -1 \end{cases}$.

$$\int_{-2}^2 f(x) dx = \int_{-2}^{-1} \frac{1}{x} dx + \int_{-1}^2 3 dx = \ln|x| \Big|_{-2}^{-1} + (3x) \Big|_{-1}^2$$

$$= [\ln|1| - \ln|-2|] + [3 \cdot (2) - 3 \cdot (-1)] = -\ln 2 + 9$$